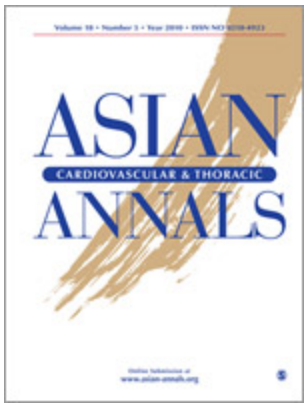


Intraoperative Acute Airway Obstruction by Ruptured Hydatid Cyst Membranes During Pulmonary Cystectomy: A High-Stakes Rescue Management

Journal:	*Asian Cardiovascular and Thoracic Annals*
Manuscript ID	Draft
Manuscript Type:	Case Study (THORACIC)
Keywords:	Echinococcus granulosus, Airway Obstruction, Double-Lumen Endotracheal Tube, Fiberoptic Bronchoscopy, Case Report
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ABSTRACT

Pulmonary hydatid disease remains a significant global public health concern in endemic regions. While surgical intervention via cystectomy and capitonnage is the definitive treatment of choice, intraoperative rupture and subsequent bronchial spillage of laminated membranes pose a catastrophic risk of complete airway obstruction and severe hypoxemia. We report a high-stakes rescue management of a 33-year-old female with immune thrombocytopenia and necrotizing pneumonia secondary to a ruptured pulmonary hydatid cyst. During elective left-sided double-lumen tube ventilation for a right pulmonary cystectomy, complete failure of right lung expansion occurred upon converting to two-lung ventilation. Immediate fiberoptic bronchoscopy revealed total occlusion of the right main bronchus by hydatid membranes. Rapid airway management utilizing a laryngeal mask airway for baseline ventilation, followed by single-lumen endotracheal intubation and urgent rigid-like fiberoptic bronchoscopic extraction by pulmonologists, successfully cleared the airway. This case underscores the critical need for meticulous interdisciplinary coordination between thoracic surgery, anesthesia, and pulmonology to navigate life-threatening airway emergencies in complicated hydatidosis.

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INTRODUCTION

Pulmonary hydatid disease remains a substantial socioeconomic and biomedical challenge globally, maintaining high endemicity across regions with livestock farming such as the Mediterranean basin, the Middle East, South America, and Central Asia[1]. The condition is a zoonotic infestation caused predominantly by the larval stage of the cestode *Echinococcus granulosus*, wherein humans act as accidental intermediate hosts via the fecal-oral ingestion of tapeworm eggs. Once the embryos penetrate the intestinal mucosa and bypass the hepatic filter, they lodge within the highly vascular, compliant parenchyma of the lungs. Over time, these slow-growing lesions develop a characteristic structural architecture composed of a host-derived fibrous adventitium, an acellular laminated cuticular membrane, and an inner germinal epithelial layer that produces immunogenic hydatid fluid and protoscolices [1].

While intact pulmonary hydatid cysts are frequently diagnosed incidentally during routine imaging, structural complications such as spontaneous or iatrogenic rupture occur in a significant proportion of patients. Ruptured cysts trigger localized pericystic pneumonitis, necrotizing parenchymal destruction, or fistulization into the bronchial tree, manifesting clinically with fever, severe cough, hemoptysis, and the pathognomonic expectoration of salty cyst contents known as hydatid vomica [1].

Surgical intervention, primarily aiming for maximum parenchymal preservation through techniques such as cystectomy or capitonnage, remains the absolute mainstay of definitive treatment. However, the perioperative management of a complicated or already ruptured hydatid cyst presents severe technical difficulties and high clinical stakes for both thoracic surgeons and anesthesiologists. The standard anesthetic approach relies heavily on achieving absolute lung isolation via double-lumen endotracheal tubes (DLTs) or bronchial blockers. This

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isolation is critical to restrict the highly antigenic cyst fluid and debris to the ipsilateral side, thereby shielding the dependent, healthy contralateral lung from chemical pneumonitis, profound bronchospasm, and potentially fatal systemic anaphylactic reactions [1].

Despite precise anatomical isolation during the pre-procedural phase, the intraoperative environment remains dynamic and unpredictable. Manipulations of a fragile, chronically infected lung, fluctuations in mechanical ventilation pressures, or sudden alterations in patient positioning can shift hidden endobronchial fragments of the laminated membrane. If these robust, gelatinous membranes migrate or collapse into the central airways, they can precipitate sudden, catastrophic mechanical asphyxia by totally impacting major bronchial orifices. While the existing thoracic literature extensively details the surgical techniques of cyst removal, concrete management algorithms for acute, massive airway impaction during the transition between one-lung and two-lung ventilation remain poorly standardized.

Herein, we report a high-stakes multidisciplinary rescue management of a 33-year-old female patient with pre-existing immune thrombocytopenia (ITP) who experienced a sudden, life-threatening right main bronchus occlusion by hydatid cyst membranes during an elective pulmonary cystectomy. This case outlines the crucial role of rapid clinical decision-making and interdisciplinary synergy between anesthesia, thoracic surgery, and pulmonology to successfully navigate unexpected endobronchial disasters.

CASE DESCRIPTION

A 33-year-old female (weight: 60 kg, height: 175 cm) with a known history of immune thrombocytopenia (ITP) and hydatid disease was referred to our department for a planned pulmonary cystectomy. She had presented to the emergency department one month prior with

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fever and hemoptysis. A baseline chest computed tomography (CT) scan revealed extensive consolidation with central necrosis in the right upper lobe extending to the apex, alongside centrilobular nodules and a tree-in-bud appearance, initially suggestive of necrotizing pneumonia. It also noted a slightly prominent 2-mm bronchial arterial branch originating from the descending thoracic aorta. Subsequent bronchoscopy and bronchoalveolar lavage (BAL) by pulmonologists revealed membranous structures in the right upper and lower lobes, which pathology confirmed as a hydatid cyst with neutrophil-rich exudate. The patient received a complete preoperative course of albendazole to reduce cyst viability prior to scheduled surgery alongside wide spectrum antibiotherapy for necrotizing pneumonia.

Upon entering the operating room, standard American Society of Anesthesiologists (ASA) monitoring was established, combined with Bispectral Index (BIS) monitoring. Intravenous induction was initiated via target-controlled infusion (TCI) of propofol and remifentanil. Following easy bag-valve-mask ventilation, rocuronium (1 mg/kg) was administered. A 35 Fr left selective DLT was placed on the first attempt using video laryngoscopy (Storz Mac 3 blade), and its precise positioning was confirmed via fiberoptic bronchoscopy (FOB). Following left radial artery and urinary catheterization, the patient was transitioned into the left lateral decubitus position, and tube placement was re-verified via FOB.

The surgical team proceeded with the planned right-sided intervention. Upon completion of the surgical procedure, mechanical parameters were adjusted to return from one-lung ventilation to two-lung ventilation. Concurrently, it was noted that the right lung completely failed to expand. Urgent FOB inspection through the DLT revealed that the entrance of the right main bronchus was entirely impacted and occluded by hydatid cyst membranes.

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Standard endobronchial suction catheters and the thin-caliber fiberoptic bronchoscope (FOB) passed through the 35 Fr double-lumen tube (DLT) failed to evacuate or dislodge the material. Given the narrow internal diameter of each individual lumen in a 35 Fr DLT (approximately 3.5 to 4.0 mm), suctioning or retrieving thick, tenacious, and highly elastic gelatinous hydatid membrane fragments through the tube was technically impossible. Because the DLT lumens themselves had become mechanically impacted, blocking both ventilation and bronchoscopic access, immediate extubation of the DLT was unavoidable to restore airway patency. Upon extubation, the tube was found to be fully packed with thick hydatid cyst fragments (Figure 1). A laryngeal mask airway (LMA) was immediately inserted to secure emergency oxygenation and protect baseline ventilation, while the surgical team rapidly closed the thoracotomy field.

The patient was immediately turned back to the supine position. Using video laryngoscopy, the trachea was re-intubated with a size 8.0 single-lumen endotracheal tube. At this stage, the patient was exclusively dependent on single-lung ventilation of the left lung. The pulmonology team was urgently mobilized to the operating room; they advanced a fiberoptic bronchoscope through the endotracheal tube and systematically extracted the massive, impacted hydatid mass obstructing the right main bronchus (Figure 2). Immediately following the clearance of the obstruction, full expansion of the right lung was achieved, and the right-sided chest tube began to oscillate normally.

Neuromuscular blockade was reversed with sugammadex (6 mg/kg IV), and the patient was successfully extubated in the operating room. Showing no signs of respiratory distress, she was transferred to the cardiovascular surgery intensive care unit (ICU) under monitored oxygen supplementation (5 L/min via a simple face mask). Her ICU stay lasted two days, after which she was safely transferred to the regular ward in stable condition.

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Following a routine thoracic surgery follow-up process, the patient's chest tube was removed on the 4th postoperative day. After spending an additional 2 days in the ward, the patient was discharged without further complications. At the tenth day post-discharge follow-up, the chest radiograph demonstrated a fully expanded lung. The patient presented with no new symptoms and had no complaints other than mild pain. While albendazole therapy was ongoing, long-term imaging and hematological evaluations were scheduled by the thoracic surgery and infectious diseases teams.

DISCUSSION

The primary therapeutic cornerstone for managing both complicated and uncomplicated pulmonary hydatid cysts is surgical resection, aimed at eradicating the parasite, closing bronchial communications, treating the residual cystic bed, and preventing recurrence while preserving the maximum volume of functional lung parenchyma. However, as demonstrated in our patient, the perioperative path of a ruptured, chronically infected pulmonary hydatid cyst is highly volatile and poses extraordinary challenges for the multidisciplinary team [1]. Complicated cysts, often presenting as necrotizing consolidations, severe pericystic pneumonitis, or lung abscesses, are associated with a substantial increase in postoperative morbidity, which ranges from 6.5% to 39% in the literature [1].

In endemic regions, establishing a high index of clinical suspicion remains crucial for timely diagnosis and surgical planning [1]. While advanced imaging like computed tomography (CT) and magnetic resonance imaging (MRI) provides crucial anatomical detail regarding the extent of parenchymal damage and multi-organ involvement, serological tests such as ELISA can occasionally yield false-negative results, particularly if antigen release into the bloodstream is

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restricted [1-3]. When a cyst ruptures into the bronchial tree, direct microscopic examination or cytopathological analysis of sputum or bronchoalveolar lavage (BAL) fluid provides a definitive 100% diagnostic confirmation by identifying classic laminated cuticle fragments, hooklets, or protoscolices within a neutrophil-rich exudate [1]. To minimize the risk of intraoperative parasite dissemination, systemic anaphylaxis, or post-surgical recurrence, initiating preoperative antiparasitic treatment with albendazole (10–15 mg/kg/day) is strongly recommended to reduce cyst viability and structural tension [1, 4].

The anesthetic management of thoracic hydatidosis mandates absolute lung isolation, typically achieved through the placement of a double-lumen endotracheal tube (DLT) [1, 5, 6]. This strategy is critical to isolate ventilation, facilitate surgical maneuvers, allow continuous endobronchial suctioning, and crucially, protect the healthy contralateral lung from the catastrophic spillage of highly antigenic hydatid fluid and obstructive membranes [1, 5, 6]. However, the intraoperative environment remains dynamic and unpredictable. Patient positioning, such as transitioning to the lateral decubitus position, combined with alterations in mechanical ventilation and positive airway pressures, can cause fragmented, floating endobronchial membranes to shift or impact into the central airways [5, 6]. In our case, although the left-sided DLT initially provided effective isolation, the unexpected migration of elastic membranes into the DLT lumens led to complete tube occlusion and acute inability to ventilate. While converting to an LMA temporarily compromised lung isolation—representing a key limitation in standard airway strategy—it provided an indispensable emergency ventilatory bridge during rapid surgical closure. This scenario highlights a critical clinical dilemma: when narrow DLT lumens become mechanically occluded by solid hydatid debris, conventional suctioning via FOB often fails. In such high-stakes emergencies, rapid conversion to a large-

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bore single-lumen tube (size 8.0) is required to allow therapeutic clearance using larger fiberoptic or rigid instruments, even if it entails a transient loss of isolated lung protection.

Historically, managing the intraoperative escape of solid hydatid debris into the tracheobronchial tree has relied heavily on immediate rigid bronchoscopy to scoop out impacted fragments, using the rigid scope itself as a temporary tracheal tube to maintain airway patency and isolate the lungs [6]. Conversely, in non-surgical or chronic settings, minimally invasive techniques have evolved. For instance, Hejazi et al. described a novel flexible bronchoscopic "saline injection method" where hypertonic saline is delivered through a thin catheter to detach adherent, ruptured hydatid membranes from the mucosal bronchial wall, achieving successful non-surgical extraction without general anesthesia [4]. In our high-stakes intraoperative scenario, emergency flexible bronchoscopy performed through the single-lumen tube functioned seamlessly as a rigid-like extraction tool, completely clearing the impacted right main bronchus and immediately restoring lung expansion and normal chest tube oscillation.

The choice of surgical approach for thoracic hydatid disease continues to be a subject of discussion. Traditional posterolateral thoracotomy remains the safest and most widely utilized incision, providing excellent exposure for large or complicated cysts and immediate control of hilar structures [1]. For bilateral pulmonary hydatid disease, historical protocols favored a staged bilateral thoracotomy, however, extensive institutional experience, such as the landmark series of 60 consecutive patients by Çetin et al., demonstrated that a single-stage median sternotomy significantly reduces total hospitalization time (17.5 days versus 47.3 days for two-stage approaches) and minimizes pain-related pulmonary complications like secretion retention and atelectasis [7]. Indeed, with contemporary advancements in surgical techniques, technological equipment, and surgical expertise, length of hospital stay has decreased

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dramatically. More recently, video-assisted thoracoscopic surgery (VATS) has gained widespread acceptance for peripheral, uncomplicated cysts, offering reduced postoperative pain and accelerated recovery [1, 2, 8]. Yet, the presence of severe inflammatory adhesions or a ruptured cyst frequently necessitates an intraoperative conversion from VATS to open thoracotomy to ensure safe, complete pericystectomy and prevent uncontrolled spillage [8]. Furthermore, multi-organ involvements, such as simultaneous pulmonary-splenic hydatidosis or mediastino-aortic cysts, frequently require complex single-stage transthoracic routes, thoraco-phrenotomy, or extensive prosthetic reconstructions [3, 9].

Fortunately, our patient's parenchymal destruction was manageable via a conservative cystectomy, avoiding radical resections like lobectomy or pneumonectomy, which are typically reserved for lungs extensively destroyed by chronic abscesses, severe bronchiectasis, or friable adventitia with multiple complex bronchial fistulas. Following successful airway clearance and surgical closure, the administration of sugammadex (6 mg/kg) ensured rapid, complete reversal of neuromuscular blockade, allowing immediate extubation in the operating room and minimizing the risks of postoperative atelectasis.

CONCLUSION

In conclusion, while preoperative medical therapy and standard lung isolation devices are vital, they do not completely eliminate the mechanical risk of acute airway impaction from loose, fragmented laminated membranes. This case demonstrates that successful survival of an acute endobronchial hydatid disaster depends entirely on rapid clinical adaptation, immediate access to fiberoptic bronchoscopic equipment, and tight interdisciplinary synergy between anesthesia, thoracic surgery, and pulmonology.

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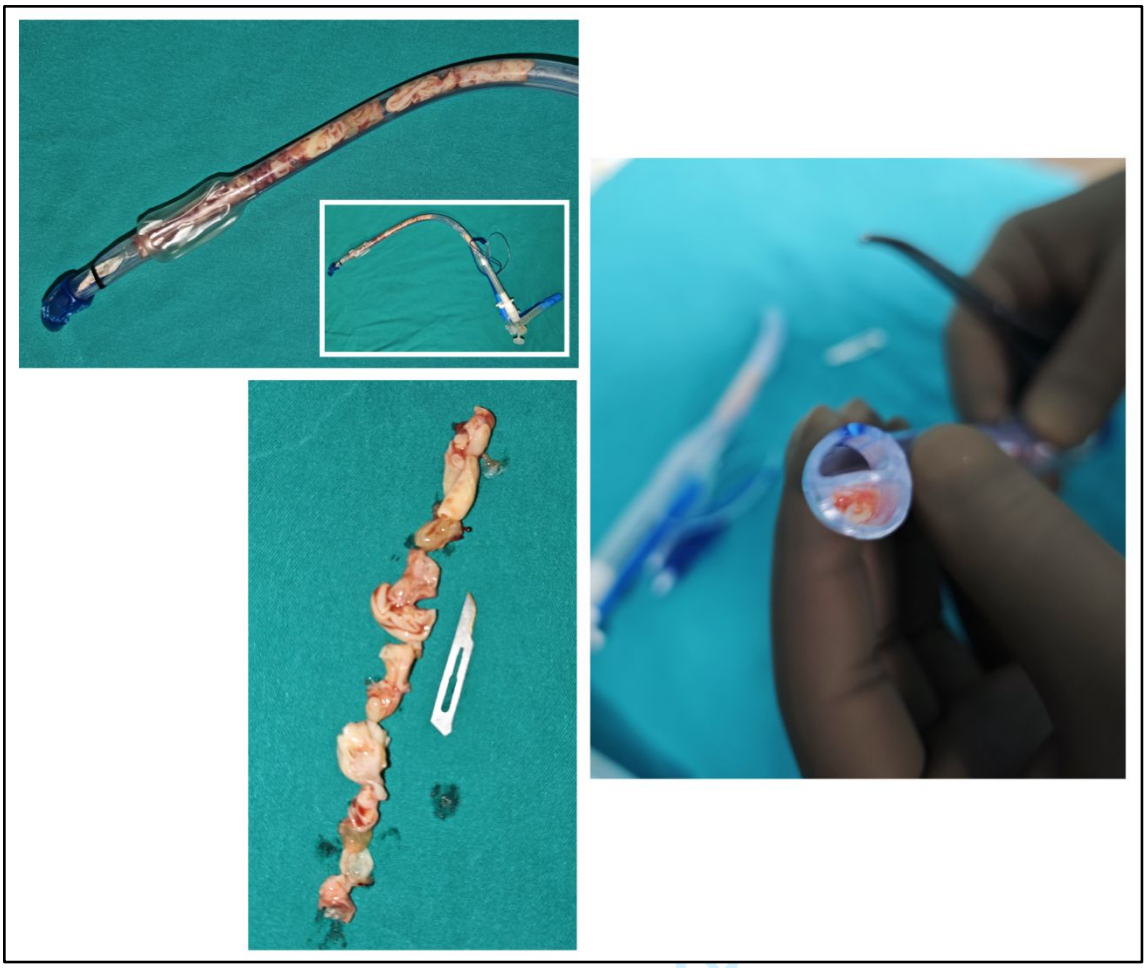


Figure 1: Fragmented pieces of the thick, whitish hydatid laminated membrane extracted immediately upon the removal of the occluded double-lumen endotracheal tube.

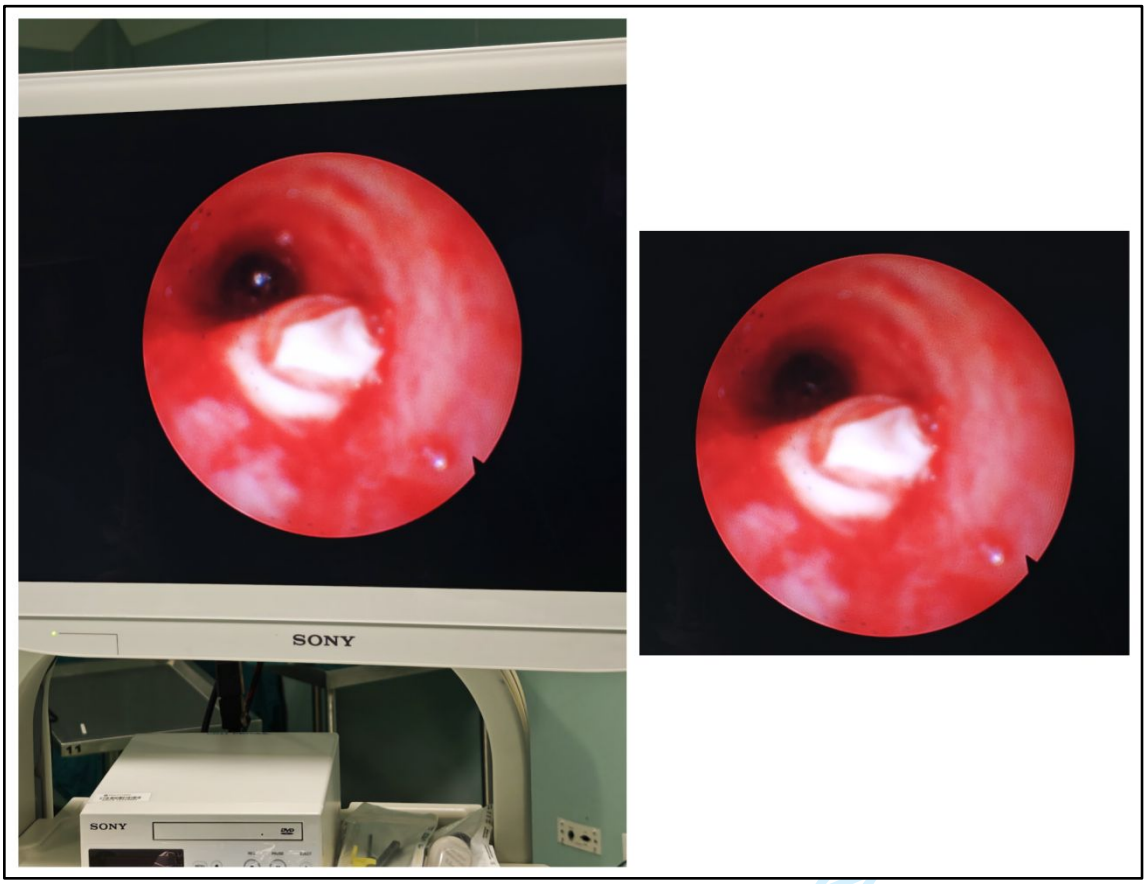


Figure 2: Endobronchial view of the impacted hydatid mass obstructing the right main bronchus via fiberoptic bronchoscopy.